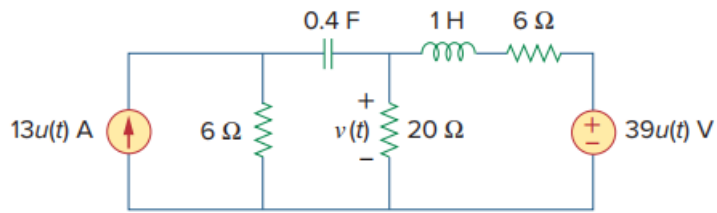


Problem

Obtain $v(t)$ for 0PSpice or *MultiSim*.

Figure 8.116:



Step-by-step solution

Step 1 of 3

Refer to Figure 8.116 in the textbook.

Consider the step function.

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Step 2 of 3

Consider the pulse source for the given input source using Pspice.

Use VPWL for pulse voltage source, and IPWL for the pulse current source.

Construct the circuit using PSpice as shown in Figure 1.

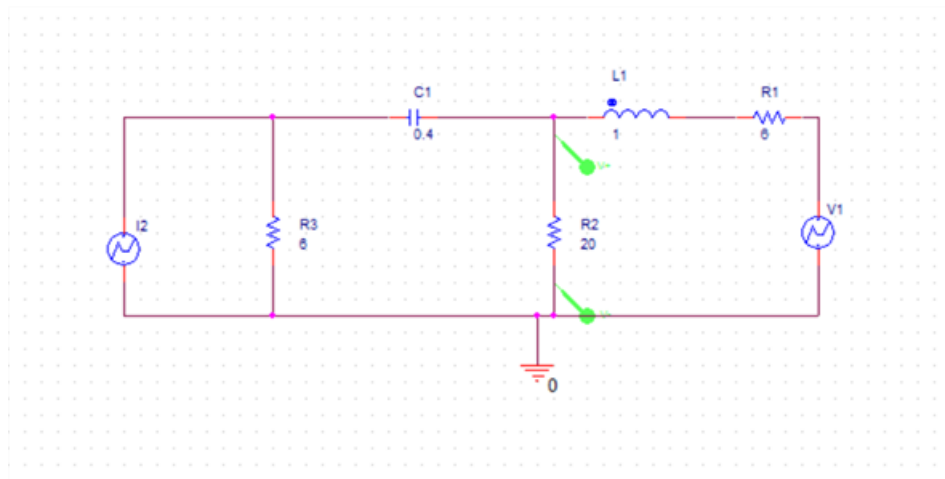


Figure 1

Step 3 of 3

The response for $v(t)$ for $0 < t < 4$ is shown in Figure 2.

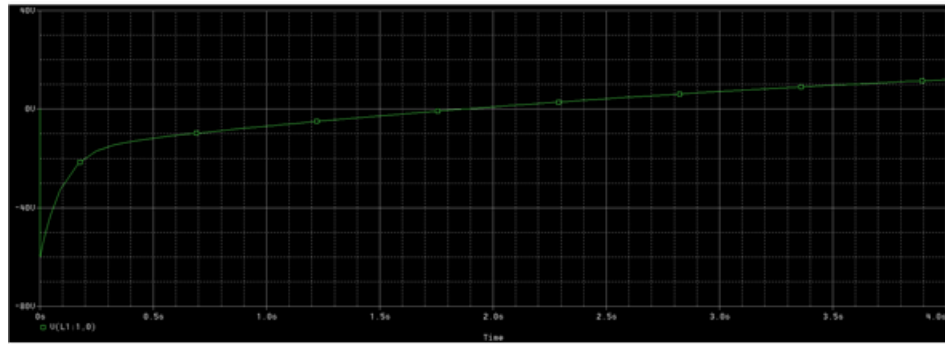


Figure 2

Hence, the voltage $v(t)$ for $0 < t < 4$, has been obtained using Pspice in Figure 2.